INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD

MA1220 - Calculus - II

Final Exam 2026

- Electronic devices are **NOT allowed**.
- No specific charts, graphs, log tables, calculators, etc. are required.
- Answer all questions with proper justification. The examination is for 40 **marks** and the duration is 2 **hours**.
- **Please start each question on a new page.**

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1. ($5 \times 1 = 5$ Marks)
   (a) Let f be a bounded function on the closed interval $[a, b]$. Define the Riemann integral $\int_a^b f(x)\,dx$.
   (b) Let f be a continuous function on $[a, \infty)$. Define the improper integral $\int_a^\infty f(x)\,dx$.
   (c) State the ϵ-δ definition of continuity of a function $f : \mathbb{R}^2 \to \mathbb{R}$ at a point $(a, b) \in \mathbb{R}^2$.
   (d) Define the directional derivative of a function $f : \mathbb{R}^2 \to \mathbb{R}$ at a point $(a, b) \in \mathbb{R}^2$ in a given direction.
   (e) Define local maxima and local minima of a function $f : \mathbb{R}^2 \to \mathbb{R}$.

2. $(2 + 4 = 6$ Marks)
   (a) Express the integral $\int_0^2 x^3\,dx$ as the limit of a Riemann sum using right endpoints. Evaluate the resulting limit.
   (b) Let $f : [0, 1] \to \mathbb{R}$ be a positive continuous function. If $\lim_{n\to\infty} \left(f\left(\frac{1}{n}\right) f\left(\frac{2}{n}\right) \cdots f\left(\frac{n}{n}\right)\right)^{\frac{1}{n}} = e^{g(x)}$, then find the function $g(x)$.

3. $(3 + 3 = 6$ Marks)
   (a) Find the length of the curve $y = \frac{x^2}{2} - \frac{\ln x}{4}, \quad 1 \le x \le 2$.
   (b) Find the area enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.

4. (3 + 3 = 6 Marks) Test the convergence/divergence of the improper integrals

$$\int_1^\infty \frac{\sin x \cos x}{x}\,dx \qquad \text{and} \qquad \int_1^\infty \frac{|\sin x \cos x|}{x}\,dx.$$

5. (2 + 2 = 4 Marks) Define the function

$$f(x,y) = \begin{cases} \dfrac{(x+2y)^2}{x+y}, & \text{if } x+y \neq 0, \\ 0, & \text{if } x+y = 0. \end{cases}$$

(a) Evaluate, using the definition of partial derivatives, $\dfrac{\partial f}{\partial x}(0,0)$ and $\dfrac{\partial f}{\partial y}(0,0)$.

(b) Is $f(x,y)$ continuous at $(0,0)$?

6. (6 marks) The temperature at a point (x,y,z) is given by

$$T(x,y,z) = 5e^{-2x^2-y^2-2z^2},$$

where T is measured in centigrade and x, y, z in meters.

(a) Find the rate of change of temperature at the point $P(1,2,-1)$ in the directi toward $(1,1,0)$.

(b) In which direction does the temperature decrease most rapidly?

(c) Find the maximum rate of decrease at P.

7. (3 Marks) Suppose that $f(x,y,z)$ is a function of three variables and let

$$\mathbf{u} = \frac{1}{\sqrt{6}}(1,1,2), \qquad \mathbf{v} = \frac{1}{\sqrt{3}}(1,-1,-1), \qquad \mathbf{w} = \frac{1}{\sqrt{3}}(1,1,1).$$

Suppose that at a point (a,b,c),

$$D_{\mathbf{u}}f = 0, \qquad D_{\mathbf{v}}f = 0, \qquad D_{\mathbf{w}}f = 4.$$

Find ∇f at (a,b,c).

8. (4 Marks) Use the method of Lagrange multipliers to find the maximum and minimu values of

$$f(x,y) = xy^2$$

on the circle

$$x^2 + y^2 = 3,$$

and the points where they occur. Explain why the extreme values exist.

All the best